

NETWORK SLICING IN SATELLITE SYSTEMS FOR RAN

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Abstract

5G and next-generation 6G networks are challenging by the rapid evolution of 5G, allowing more flexible and service-oriented network ensembles to be developed to host heterogeneous applications that require varied Quality of Service (QoS). Network slicing is an important technology in providing coexistence of numerous virtual networks on a common physical infrastructure.

Satellite systems have been added to the current 3GPP standards as NTN to complement terrestrial cellular systems. This article describes a literature study of network slicing in satellite-based Radio Access Networks (RANs) concentrating mainly on 3GPP RAN architecture and slice realization in Low Earth Orbit (LEO) satellite systems. Such a solution shows how satellite RAN resources can be logically divided to ensure provision of layered services that would include enhancements in the form of eMBB (enhanced mobile broadband), ultra-reliable low-latency communications (URLLC), and massive Internet of Things (mIoT).

Index Terms —Network slicing, satellite RAN, 5G, 3GPP, NTN, LEO satellites.

I. Introduction

The rapid growth of mobile communication services and the rise of data-intensive and latency sensitive applications place an unrelenting strain on legacy network infrastructures. Fifth-Generation (5G) and future sixth generation (6G) networks are being designed to solve these issues through virtualization approaches, flexibility, and service-aware networking approaches. An important enabling technology in this regard is the network slicing technique that utilizes multiple logical networks over a physical infrastructure. So, Satellite

communication systems are one of the key keys for achieving worldwide connectivity especially for rural, remote and underserved regions. Satellite communication has been made an integral part of the 5G environment with Release 17 introducing 3GPP Non-Terrestrial Networks (NTN). This paper provides an exploration of network slicing in satellite-based RANs, where network slicing is of particular interest in this study, in particular, network slicing in satellite-based RAN network with regards to satellite RANs using 3GPP and describes the framework for LEO satellite slicing, and studies how 3GPP-based satellite systems can be constructed as the foundation for network slicing based on 3GPP standardization for achieving this goal in Low Earth Orbit (LEO).

II. Background and motivation

A. Satellite Communication Challenges

The main satellite networks were built primarily to deliver fixed broadband and broadcast services. Nonetheless, for the 5G RAN, many challenges such as limited spectrum distribution, dynamic satellite motion, and very restricted latency make the use of satellites into the scenario more complex.

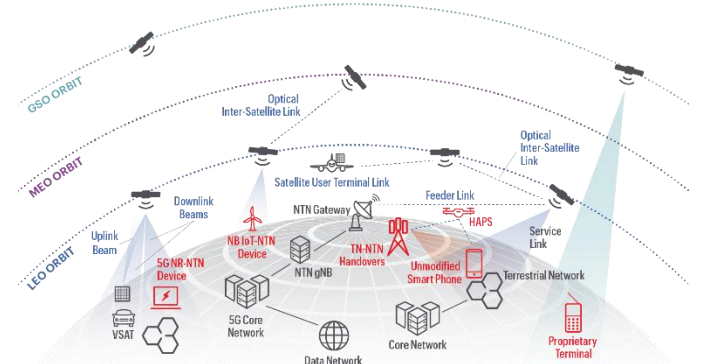


Figure: Integration of Low Earth Orbit (LEO) satellites into the 5G RAN as part of NTN

B. Motivation for Network Slicing

Network slicing is a flexible method for accommodating mixed 5G services over a host infrastructure and is based on the principle that we have no single target. With logical isolation, network slicing allows organizations to tune their resources and improve the efficiency of their spectrum as a whole, guaranteeing service contracts (SLAs) are fulfilled, even in resource-constrained satellite networks.

III. 3GPP RAN Architecture Overview

A. NG-RAN Architecture

As per 3GPP specifications, The Next Generation Radio Access Network (NG-RAN) comprises of user equipment (UEs) interlinked to 5G gNodeBs (gNBs), which directly communicate with 5G Core Network (5GC). The gNB is divided in terms of function into 3 major components including Central Unit (CU), Distributed Unit (DU), and Radio Unit (RU).

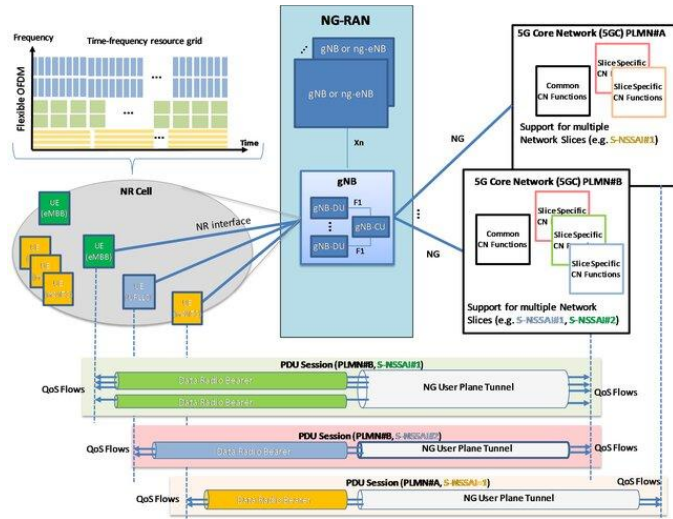


Figure: NG-RAN architecture with support for network slicing and RAN sharing

B. Functional Splits

Higher-layer protocols like Radio Resource Control (RRC) and Packet Data Convergence Protocol (PDCP) are implemented by CU, alongside the real-time scheduling of Medium Access Control (MAC) and Physical (PHY) layer are handled by DU. The

RU performs radio frequency functions. This disaggregated structure makes for virtualization and slice-aware computation possible.

IV. Satellite-based RAN Architecture

A. Satellites inclusion in RAN

Satellites, which could be found in satellite-based RAN, may act as the radio relay node, or as payloads that hold the RAN. This allows for end-to-end service continuity through ground gateways connecting the satellite network to the 5G Core.

B. LEO Satellite Characteristics

LEO satellites fly at an altitude of hundreds to thousands of km and thus have lower latency than GEO satellites. These can span large constellations, which also provide better coverage and capacity—thus, they are suited for RAN-level slicing.

V. Network Slicing Fundamentals

A. Concept of Network Slicing

Network slicing concept creates several logical networks, or slices, from a single physical network. Each slice is fit for the type of service and offers different QoS considerations (latency, throughput, and reliability).

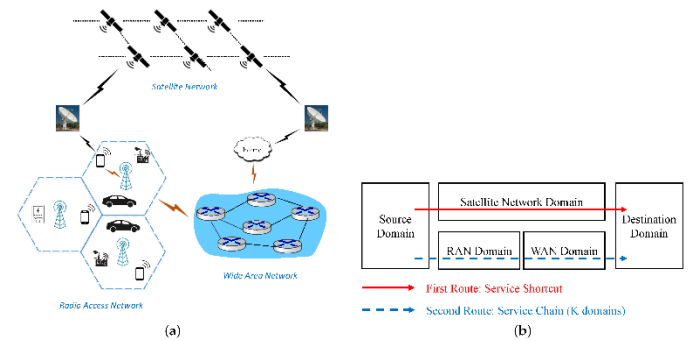


Figure: The basic structure of the multi-domain network slicing service provision market.

(a) The physical infrastructure; and (b) the logical structure.

B. Slice Types

The most popular types of slices will be enhanced Mobile Broadband(eMBB), Ultra-Reliable Low-Latency Communications (URLLC), or massive Machine-Type Communications (mMTC). Each slice is independent of the other slices and operates without the need for other slices to vary in

performance.

VI. Network Slicing in Satellite RAN

A. Slice Implementation at RAN Level

In satellite RANs, slicing is realized mainly on the MAC and PHY layers as slice-aware schedulers. Radio resources such as bandwidth, time slots, and transmission power are dynamically allocated based on slice priority and service requirements.

B. Resource Isolation and QoS Mapping

Slice isolation ensures that congestion or performance degradation in one slice does not impact others. QoS mapping aligns satellite RAN parameters with 5G core network requirements to maintain end-to-end service guarantees.

VII. Slice Management and Orchestration

Slice management is handled through standardized 3GPP management functions such as the Network Slice Management Function (NSMF). These entities oversee slice lifecycle operations including creation, activation, scaling, monitoring, and termination. Dynamic orchestration enables efficient adaptation to varying traffic demands and satellite visibility conditions.

VIII. Simulation and Performance Evaluation

A simulation-based approach is used to model a LEO satellite scenario with ground stations acting as access points. Multiple slices are defined with different priorities and bandwidth allocations. Performance metrics such as access duration and link availability are evaluated to demonstrate the effectiveness of network slicing in satellite RAN environments.

IX. Applications

Satellite RAN network slicing enables a wide range of applications including rural broadband access, disaster recovery communications, military and mission-critical services, and large-scale IoT deployments such as smart agriculture and environmental monitoring.

X. Challenges and Future Work

Key challenges include mobility management for fast-moving satellites, synchronization between terrestrial and satellite RANs, and slice orchestration complexity. Future research directions include AI-driven slice optimization, integration with 6G architectures, and real-time adaptive slicing strategies.

XI. Simulation and Results

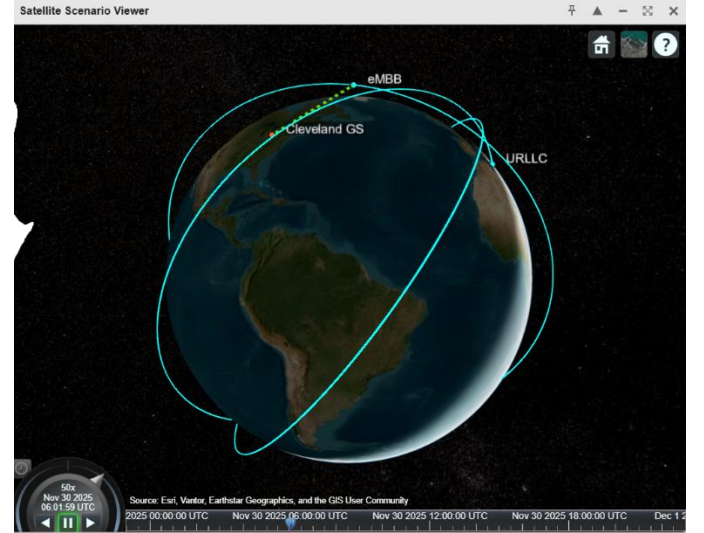
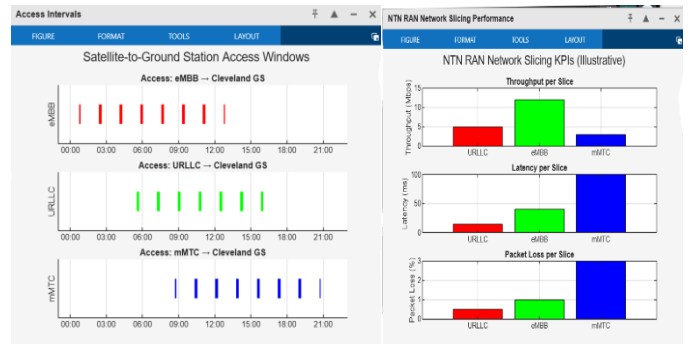


Figure: LEO satellite orbits and sliced links to Cleveland GS.

A simulation environment was developed using a LEO satellite constellation model to evaluate the feasibility of network slicing in satellite-based RAN. The simulation considers satellite-to-ground access links as RAN connectivity, where multiple logical slices are implemented through bandwidth prioritization and access scheduling. The scenario is simulated over a 24-hour period to capture satellite mobility and access dynamics.

Figure: Satellite access windows and network slicing KPIs for



eMBB, URLLC, and mMTC.

XII. Conclusion

This paper demonstrates that network slicing is a viable and effective solution for supporting diverse services in satellite-based RANs. By leveraging 3GPP-standardized architectures and LEO satellite systems, satellite networks can evolve into flexible, service-aware platforms capable of delivering 5G and beyond services with assured QoS.

XIII. References

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